

STATISTICS

PART 5: Sampling Theory (for a Single Population)

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Rev: 07.11.2024

Sampling Theory

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SAMPLING THEORY

- A **population** consists of all the members (for example, the IQ scores of all students in YTU, the weights of all people in a city) in which we are interested.
- A measure such as the mean, standard deviation, proportion, etc., that summarizes the population is called a **population parameter** or simply a **parameter**.

We can extract information from a population

1. By **full enumeration** if possible (if the population has a small number of members, it has no high cost according to time or money, and the measurement of data does not damage the products we are interested in).
2. By choosing a **sample** from it. This process is called **sampling**. A sample is a subset of the population, and it should be representative of the population. The measures (mean, standard deviation, etc.) of a sample are called the **sample statistic** or simply **statistic**.

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Sampling Methods

Non-Stochastic Methods

- Convenience Sampling
- Judgmental Sampling
- Quota Sampling
- Snowball Sampling

Stochastic Methods

- Simple Random Sampling
- Stratified Sampling
- Systematic Sampling
- Cluster Sampling

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Non-Stochastic Methods

Convenience Sampling

Members of the population are chosen based on their relative **ease of access**.

Ex: selecting shoppers at a single mall

Judgmental Sampling

Choosing a sample from a population according to **personal experiences and thoughts**.

Ex: choosing university students for market research of a new product

Quota Sampling

selecting a sample from a population that reflects **characteristics in predetermined proportions**

Ex: A quota of 50% women

Snowball Sampling

A member is selected randomly. This member refers to a contact. Then this contact refers to another contact and so on.

Ex: Choosing the people using the drug

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Stochastic Methods

Simple Random Sampling

Each member of the population is chosen randomly, ensuring that all possible samples have an **equal likelihood** of being selected.

Stratified Sampling

If the population is not homogeneous, it is divided into subgroups (**strata**) based on specific characteristics (such as age, income). A random sample is then selected from each strata.

Systematic Sampling

We randomly select a starting member from the population list, and then every ***k*th** member is chosen. Here, ***k*** is called sampling interval.

Cluster Sampling

The population is divided into **clusters**, often based on geographical location or other natural divisions. A random sample is then selected from each cluster. Here, clustering is not based on inhomogeneity within the population.

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SAMPLING DISTRIBUTION

Let a sample of size n be randomly selected from a population of size N and let $\hat{\theta}$ be a statistic of the sample. Consider all possible combinations of the samples that can be selected from this population. If their statistics are

$$\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \dots, \hat{\theta}_{\binom{N}{n}},$$

then this distribution is called the **sampling distribution of the statistic $\hat{\theta}$** .

$$\underbrace{\hat{\theta}}_{\text{random variable}} = \underbrace{\{\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \dots, \hat{\theta}_{\binom{N}{n}}\}}_{\text{possible values of the random variable}}$$

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Sampling Distribution of the Sample Mean

When all possible samples of size n are selected from a population of size N , and the mean ($\bar{X}_i$) of each sample is calculated, we obtain the **sampling distribution of the sample means**. In this case, the statistic is

$$\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{j=1}^n X_j$$

where $X_j, j = 1, 2, \dots, n$ are the members of the chosen sample, and $\bar{X}$ is the random variable of the sampling distribution of the sample means. If the samples are selected without replacement, the sampling distribution is

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_{\binom{N}{n}}.$$

If the samples are selected with replacement, then the sampling distribution is

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_{N^n}.$$

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Theorem: If the samples of size n are selected without replacement in a population of size N , the mean and standard deviation of $\bar{X}$ are

$$E(\bar{X}) = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}},$$

where μ is the mean of the population, σ is the standard deviation of the population, $\sigma_{\bar{x}}$ is the standard deviation of the sampling distribution of the sample means. If the samples are selected with replacement, then:

$$E(\bar{X}) = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

$$\sqrt{\frac{N-n}{N-1}}$$



finite population
correction factor

$$\lim_{N \rightarrow \infty} \sqrt{\frac{N-n}{N-1}} = 1$$

If $n \leq 0.05 N$, we use the formulas for the samples that be replaced back, although the samples are selected without replacement. From now on, we assume that samples are selected with replacements unless otherwise stated.

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The **proof** for the mean and standard deviation of the sampling distribution $\bar{X}$ for the samples that are selected with replacements:

$$E(\bar{X}) = E\left\{\frac{1}{n} \sum_{i=1}^n X_i\right\} = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu$$

$$Var(\bar{X}) = \sigma_{\bar{x}}^2 = Var\left\{\frac{1}{n} \sum_{i=1}^n X_i\right\} = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n} \Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

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Example 1: Let the samples with size of 3 be selected without replacing them back into a population consisting of the numbers {2, 4, 10, 12, 18}. Create the sampling distribution of the sample mean and then calculate the mean and the standard deviation of this sampling distribution.

$$\mu = \frac{2 + 4 + 10 + 12 + 18}{5} = 9.2$$

$$\sigma^2 = \frac{(2 - 9.2)^2 + (4 - 9.2)^2 + (10 - 9.2)^2 + (12 - 9.2)^2 + (18 - 9.2)^2}{5} = 32.96$$

combinations of the samples	the mean of the samples ($\bar{X}$)
2,4,10	16/3
2,4,12	18/3
2,4,18	24/3
2,10,12	24/3
2,10,18	30/3
2,12,18	32/3
4,10,12	26/3
4,10,18	32/3
4,12,18	34/3
10,12,18	40/3

$$E(\bar{X}) = \frac{\frac{16}{3} + \frac{18}{3} + \dots + \frac{40}{3}}{10} = 9.2$$

$$\sigma_{\bar{x}} = \sqrt{\frac{(\frac{16}{3} - 9.2)^2 + (\frac{18}{3} - 9.2)^2 + \dots + (\frac{40}{3} - 9.2)^2}{10}} = 2.34$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N - n}{N - 1}} = \frac{5.74}{\sqrt{3}} \sqrt{\frac{5 - 3}{5 - 1}} = 2.34$$

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Example 2: Let a coin is tossed and X be the number of occurring head. If X a random variable of a population,

- find the mean and standard deviation of the population.
- If the coin is tossed two times ($n = 2$), show the sampling distribution of sample means. Then, compute $E(\bar{X})$ and $\sigma_{\bar{X}}$ by using
 - sampling distribution and
 - probability function.

Solution:

a)

Tossing a coin

	X	Probability $f(X)$
Tail	0	$1/2$
Head	1	$1/2$

$$\mu = E(X) = \sum_{i=1}^2 X_i f(X_i) = 0 \frac{1}{2} + 1 \frac{1}{2} = \frac{1}{2}$$

$$var(X) = \sum_{i=1}^2 (X_i - \mu)^2 f(X_i) = \left(0 - \frac{1}{2}\right)^2 \frac{1}{2} + \left(1 - \frac{1}{2}\right)^2 \frac{1}{2} = \frac{1}{4}$$

$$\Rightarrow \sigma_X = \sigma = \frac{1}{2}$$

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Solution:

b)

Tossing a coin 2 times ($n = 2$)		
tossing 1, tossing 2	Probability $f(\bar{X})$	$\bar{X}$
Head-Head ($x = 1, x = 1$)	$1/4$	1
Head-Tail ($x = 1, x = 0$)	$1/4$	0.5
Tail-Head ($x = 0, x = 1$)	$1/4$	0.5
Tail-Tail ($x = 0, x = 0$)	$1/4$	0

i) by sampling distribution

$$\bar{X} = \{0, 0.5, 0.5, 1\}$$

$$E(\bar{X}) = \frac{0 + 0.5 + 0.5 + 1}{4} = \frac{1}{2} = 0.5$$

$$\sigma_{\bar{X}} = \sqrt{\frac{(1 - 0.5)^2 + (0.5 - 0.5)^2 + (0.5 - 0.5)^2 + (0 - 0.5)^2}{4}} = \sqrt{\frac{1}{8}}$$

ii) by probability function

$$X = \{0, 0.5, 1\}$$

$$E(\bar{X}) = \sum_{i=1}^3 \bar{X}_i f(\bar{X}_i) = 1 \frac{1}{4} + 0.5 \frac{1}{2} + 0 \frac{1}{4} = \frac{1}{2} = 0.5$$

$$var(\bar{X}) = \sum_{i=1}^3 (\bar{X}_i - E(\bar{X}))^2 f(\bar{X}_i) = \frac{1}{4} \frac{1}{4} + 0 \frac{1}{2} + \frac{1}{4} \frac{1}{4} = \frac{1}{8}$$

$$\Rightarrow \sigma_{\bar{X}} = \sqrt{1/8}$$

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Theorem: If a population has a normal distribution with mean μ and standard deviation σ , then the random variable

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

Proof: Let the sample be $\{X_1, X_2, \dots, X_n\}$. If the selection is made with replacement, then $X_1, X_2, \dots, X_n$ are independent. Since

$$X_i \sim N(\mu, \sigma^2), i = 1, 2, \dots, n,$$

then

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \dots + X_n) \sim N\left(n\frac{1}{n}\mu, n\frac{1}{n^2}\sigma^2\right) = N\left(\mu, \frac{1}{n}\sigma^2\right)$$

Since $\bar{X}$ has normal distribution,

$$z = \frac{\bar{X} - E[\bar{X}]}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

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Example 3: The height of male students in a university is normally distributed with a variance of 19. If 25 male students are selected from this community, what is the probability that the difference between the sample mean and the population mean is least than 2 cm?

$$n = 25$$

$$X \sim N(\mu, 19)$$

$$\begin{aligned} P[|\bar{X} - \mu| \leq 2] &= P\left[\frac{|\bar{X} - \mu|}{\sigma/\sqrt{n}} \leq \frac{2}{\sigma/\sqrt{n}}\right] = P\left[\frac{|\bar{X} - \mu|}{\sigma_{\bar{X}}} \leq \frac{\sqrt{n} \cdot 2}{\sigma}\right] = P\left[|Z| \leq \frac{\sqrt{25} \cdot 2}{\sqrt{19}}\right] \\ &= P[|Z| \leq 2.29] = P[-2.29 \leq Z \leq 2.29] = 2(0.4890) = 0.9780 \end{aligned}$$

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Sampling Distribution of the Sample Variance

When all possible samples of size n are selected from a population of size N , and the variance (S_i^2) of each sample is calculated, we obtain the **sampling distribution of the sample variances**.

$$S_1^2, S_2^2, \dots, S_{N^n}^2$$

Here, the statistic is the sample variance S^2 , which is defined as

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

The sample standard deviation $S = \sqrt{S^2}$, and **sampling distribution of the standard deviations** are

$$S_1, S_2, \dots, S_{N^n}$$

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Theorem: If S^2 is the variance of a random sample of size n that is selected from a population which has normal distribution with variance σ^2 , the random variable

$$\chi_{n-1}^2 = \frac{(n-1)S^2}{\sigma^2}$$

has the chi-squared distribution with the degree of freedom $n-1$.

Proof:

$$\begin{aligned}(n-1)S^2 &= \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n [(X_i - \mu) - (\bar{X} - \mu)]^2 \\&= \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \underbrace{\sum_{i=1}^n (X_i - \mu)}_{n\bar{X} - n\mu} + \underbrace{n(\bar{X} - \mu)^2}_{n(\bar{X} - \mu)^2} = \sum_{i=1}^n (X_i - \mu)^2 - 2n(\bar{X} - \mu)^2 + n(\bar{X} - \mu)^2\end{aligned}$$

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$$\Rightarrow (n-1)S^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

If we divide both parts by σ^2 , then we have:

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \underbrace{\left(\frac{X_i - \mu}{\sigma}\right)^2}_{Z_i^2} - \underbrace{\left(\frac{X_i - \mu}{\sigma/\sqrt{n}}\right)^2}_{Z^2}$$

$$\Rightarrow \frac{(n-1)S^2}{\sigma^2} = \underbrace{\sum_{i=1}^n Z_i^2}_{\chi_n^2} - \underbrace{Z^2}_{\chi_1^2}$$

$$\Rightarrow \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

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Example 4: A random sample of size 16 is selected from a population that is normally distributed. If the variance of the sample is 27.6, then find the probability that the variance of the population is more than 29.

$$n = 16, S^2 = 27.6$$

$$X \sim N(\mu, \sigma^2)$$

$$\begin{aligned} P(\sigma^2 > 29) &= P\left(\frac{1}{\sigma^2} < \frac{1}{29}\right) = P\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)s^2}{29}\right) \\ &= P\left(\chi_{15}^2 < \frac{(n-1)s^2}{29}\right) = P\left(\chi_{15}^2 < \frac{(16-1)27.6}{29}\right) = P(\chi_{15}^2 < 14.3) \\ &= 1 - 0.50 = 0.50 \end{aligned}$$

Sampling Theory

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Sampling Distribution of the Sample Mean (for unknown population variance)

Theorem: If a population has a normal distribution with mean μ and unknown standard deviation σ , then the random variable

$$t_{n-1} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with the degree of freedom $n-1$.

Proof: Since $X \sim N(\mu, \sigma^2)$,

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}, \quad \chi_{n-1}^2 = \frac{(n-1)S^2}{\sigma^2}$$

From $t_v \sim \frac{z}{\sqrt{\chi_v^2/v}}$,

$$t_n \sim \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} / (n-1)}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \frac{\sqrt{n-1}}{\frac{\sqrt{n-1}S}{\sigma}} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

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Example 5: A random sample of size 16 is selected from a population that is normally distributed. If the mean and the standard deviation of the sample is 75 and 8.5, respectively, then find the probability that the mean of the population is more than 78.5.

$$n = 16, \bar{X} = 75, S = 8.5$$

$$X \sim N(\mu, \sigma^2)$$

$$\begin{aligned} & P(\mu > 78.5) \\ &= P(-\mu < -78.5) \\ &= P\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < \frac{\bar{X} - 78.5}{S/\sqrt{n}}\right) \\ &= P\left(t_{15} < \frac{75 - 78.5}{8.5/\sqrt{16}}\right) \\ &= P(t_{15} < -1.647) \end{aligned}$$

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Since the area corresponding to the value 1.647 with the degree of freedom 15 does not exist in the t table, we will use the interpolation for:

$$P(t_{15} < -1.341) = 0.10$$

$$P(t_{15} < -1.753) = 0.05$$

1.341	0.1
1.647	Y
1.753	0.05

$$\frac{1.753 - 1.341}{1.647 - 1.341} = \frac{0.05 - 0.10}{Y - 0.10}, \quad Y = 0.0629$$

$$\rightarrow P(\mu > 78.5) = 0.0629$$

Central Limit Theorem

Statistics

Sampling Distribution of the Sample Mean (for unknown distribution)

The populations are assumed to be normally distributed in the theorems given above. If the population is not normally distributed (or its distribution is unknown), we can apply the Central Limit Theorem which is given below.

The Central Limit Theorem: If a population has **any (unknown)** distribution with mean μ and standard deviation σ , then sample mean $\bar{X}$ has normal distribution for the limiting form of the sample size as $n \rightarrow \infty$.

Accordingly, the random variable

$$z = \frac{\bar{X} - E[\bar{X}]}{\sigma_{\bar{x}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has the standard normal distribution when $n \rightarrow \infty$.

Central Limit Theorem

Statistics

- In practice, if $n \geq 30$ and the distribution of the population is unknown, then the random variable

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is approximately distributed standard normally.

- If we don't know the variance of the population, we can assume that the random variable

$$z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

is distributed standard normally.

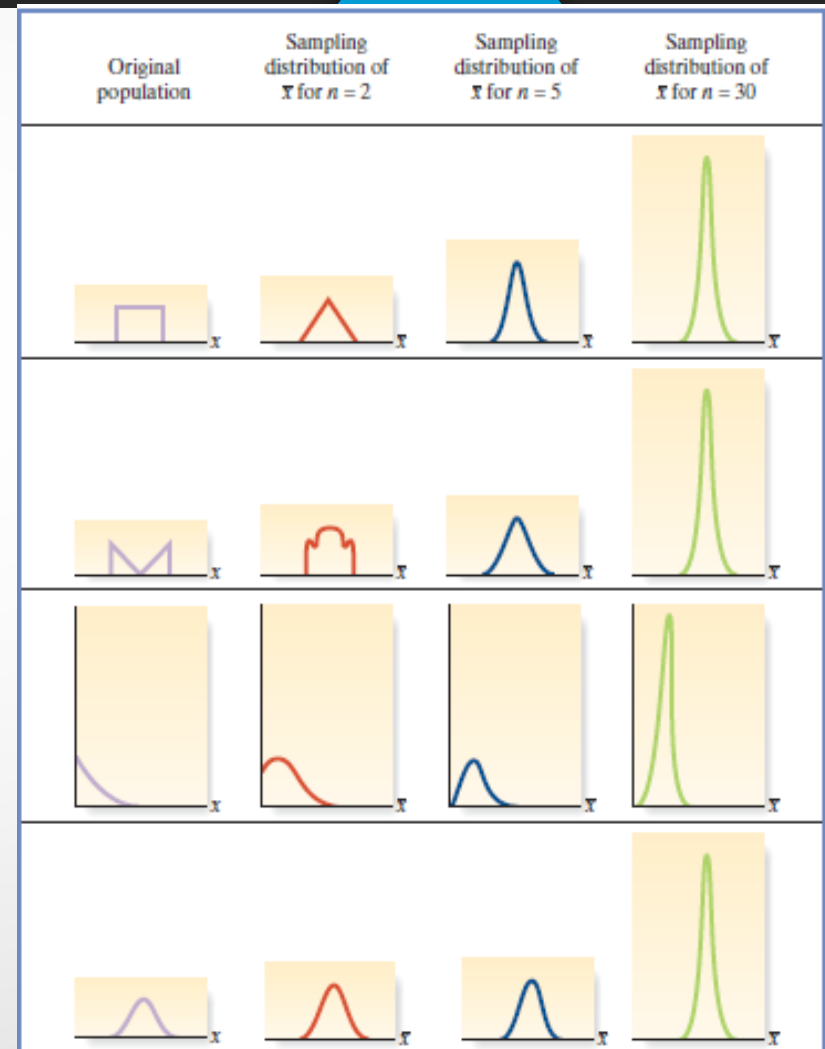
- Note that the last implication is not based on a theoretical derivation; it is just a practical approximation.

Central Limit Theorem

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As sample size (n) increases, the sampling distribution of sample means ($\bar{X}$) approximates a normal distribution.

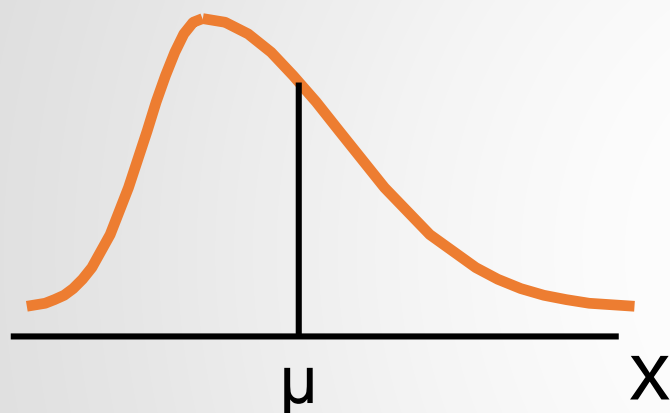
When a population has more skewness, a larger sample size is required to approximate a normal distribution.



Central Limit Theorem

Statistics

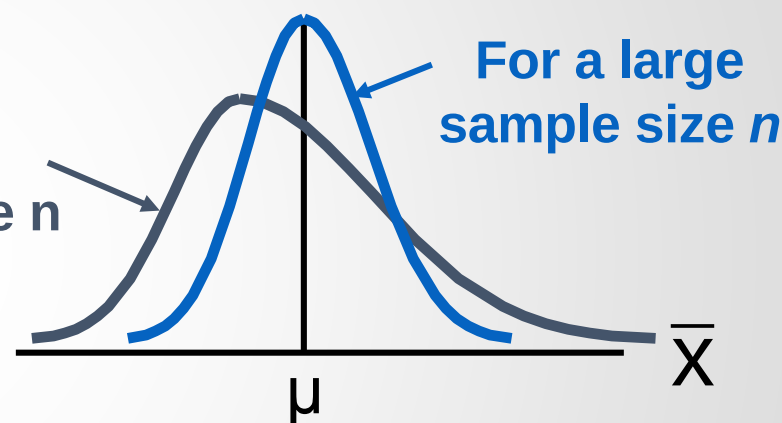
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$$E(X) = \mu$$

$$\sigma_X = \sigma$$

For a small
sample size n



$$E(\bar{X}) = \mu$$

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

As sample size (n) increases, the sampling distribution of sample means ($\bar{X}$) approximates a normal distribution with mean μ and standard deviation $\sigma / \sqrt{n}$.

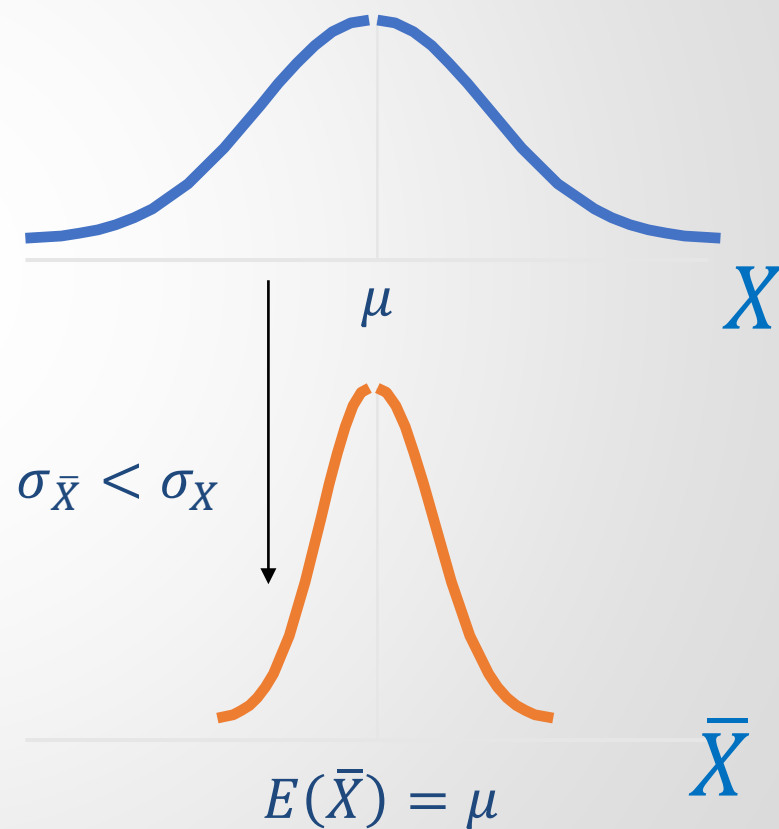
Central Limit Theorem

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Consider a population having a normal distribution with mean μ and standard deviation σ .

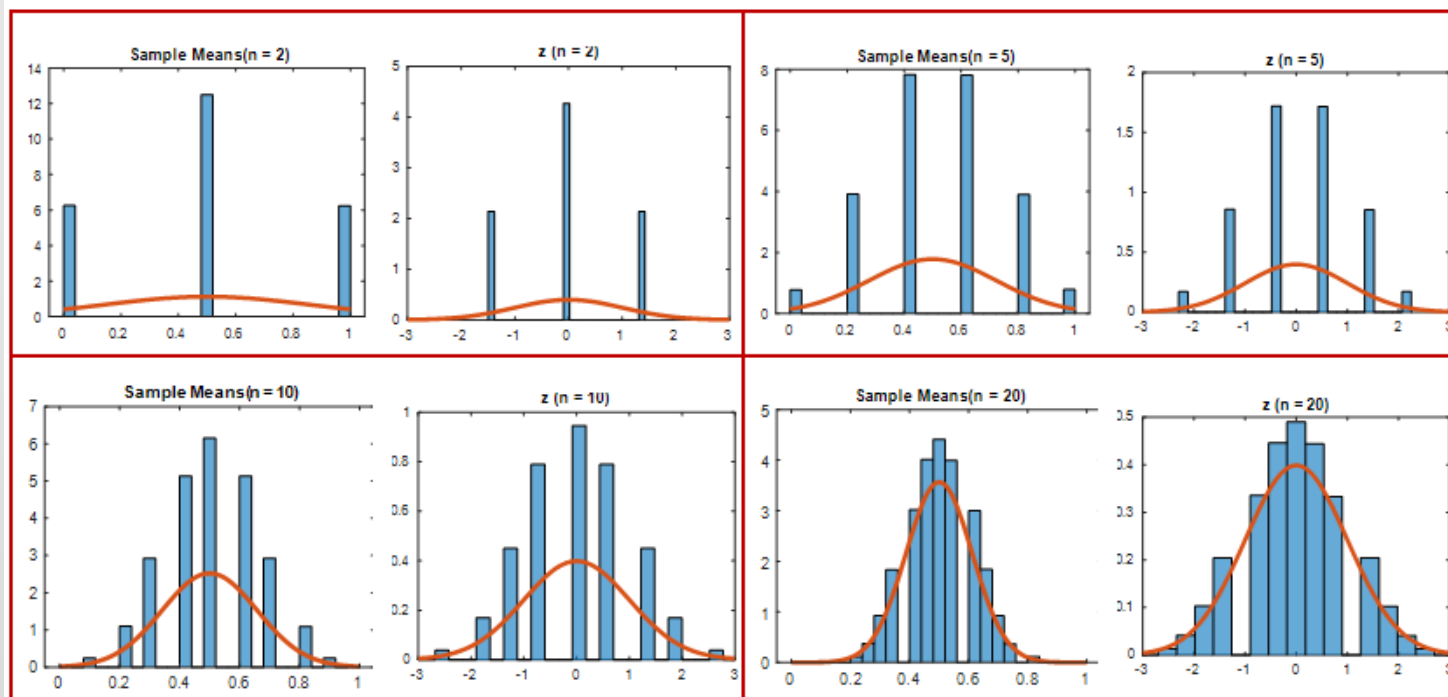
Since $\sigma_{\bar{X}} = \sigma / \sqrt{n}$,
as n increases, $\sigma_{\bar{X}}$ decreases.



Central Limit Theorem

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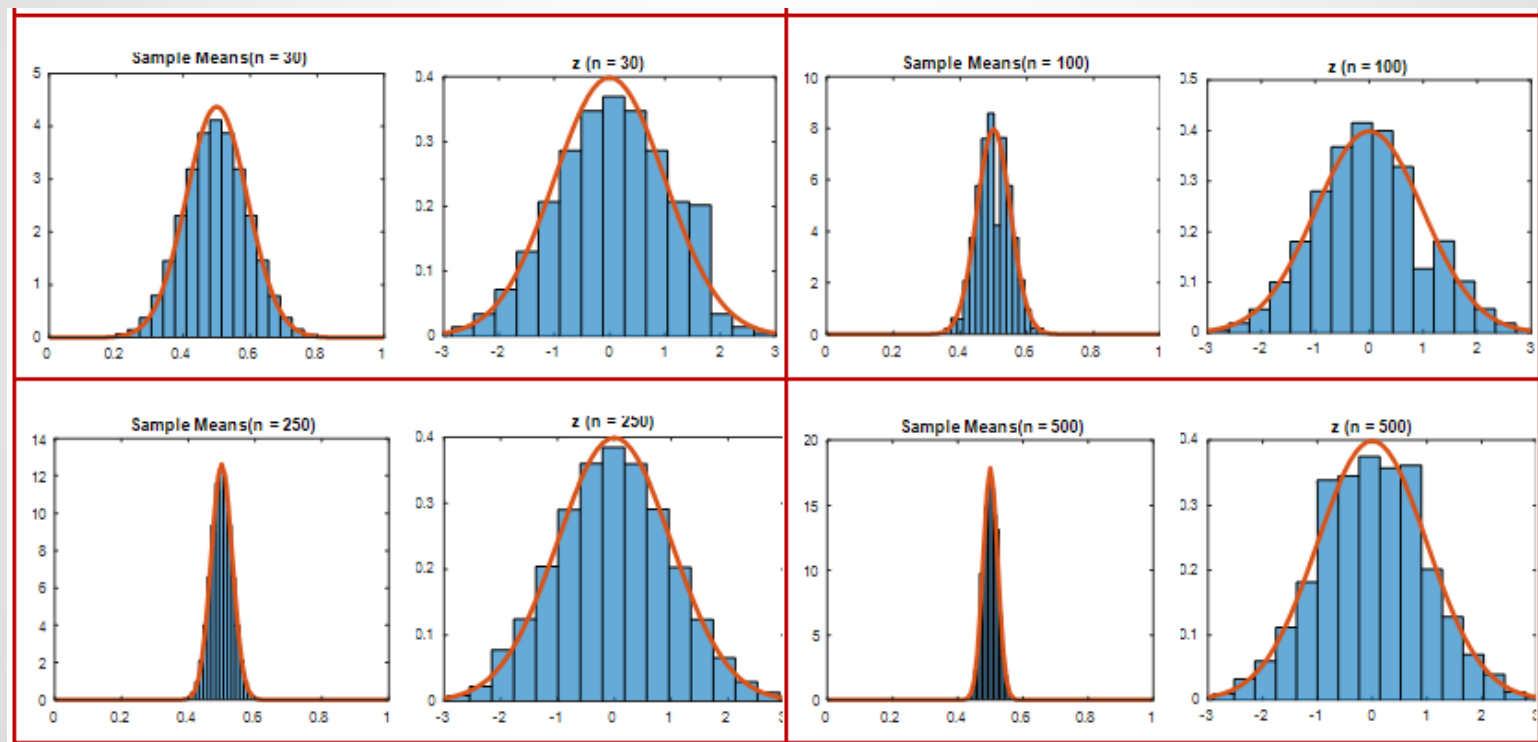
Simulation: Suppose that a coin is tossed 1 million times ($N = 1,000,000$). Tail and Head are represented by 0 and 1, respectively. When n was chosen as 2, 5, 10, 20, 30, 100, 250 and 500, the simulation produced the following histograms for sample means and their transformations to standard normal distribution. As seen, it approximates normal distribution for a large value of n . (The mean and standard deviation of the population is 0.5.)



Central Limit Theorem

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Example 6: The bottles are filled on average 75 cl soda by a machine in a factory. It is known that the standard deviation of the amount of soda in the bottles is 4.8. If a random sample of size 42 units is selected, find the probability that the sample mean is more than 77 cl.

$$n = 42, \mu = 75, \sigma = 4.8$$

$$\begin{aligned} & P(\bar{X} > 77) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{77 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(Z > \frac{77 - 75}{4.8/\sqrt{42}}\right) \\ &= P(Z > 2.7) \\ &= 0.5 - 0.4965 \\ &= 0.0035 \end{aligned}$$

Sampling Theory

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Sampling Distribution of the Sample Proportion

We know that the binomial distribution can be used for the selection with replacement from any population or for the selection without replacement from a large population. If X is a random variable for the binomial distribution, then the mean and the variance of X is

$$E(X) = np,$$

$$\text{Var}(X) = npq,$$

where p is the probability of success for a Bernoulli trial, and it is called **the population proportion**.

Since X represents the number of successes in the Binomial distribution, if we select a sample of size n from a population and the number of the members having the requested characteristics in the sample is X , then the proportion

$$\hat{p} = \frac{X}{n}$$

is called **the success proportion of the sample**.

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In this case, the sample proportion $\hat{p}$ is a sample statistic while the population proportion p is a population parameter.

The mean and the variance of the sample proportion is:

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}np = p,$$

$$Var(\hat{p}) = Var\left(\frac{X}{n}\right) = \frac{1}{n^2}Var(X) = \frac{1}{n^2}npq = \frac{pq}{n},$$

and its standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}}.$$

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Theorem: According to the Central Limit Theorem, $\hat{p}$ has normal distribution for the limiting form of the sample size as $n \rightarrow \infty$ because

$$\hat{p} = \frac{X}{n} = \frac{(Y_1 + Y_2 + \dots + Y_n)}{n} = \bar{Y}$$

where Y 's are random variable of Bernoulli distribution while $\bar{Y}$ is the mean of Y .

Then, the random variable

$$Z = \frac{\hat{p} - E(\hat{p})}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

has the standard normal distribution when $n \rightarrow \infty$.

Note that, when we try to find the population proportion p (also q) in practice, we can use $\hat{p}$ and $\hat{q}$ instead of them in the square-root expression in the theorem.

$$Z = \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}}$$

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Example 7: If a balanced die is rolled 1500 times, find the probability that the difference between the frequency proportion of occurring 5 and the probability of occurring 5 is less than 0.02.

$$n = 1500, \quad p = \frac{1}{6}, \quad q = \frac{5}{6}$$

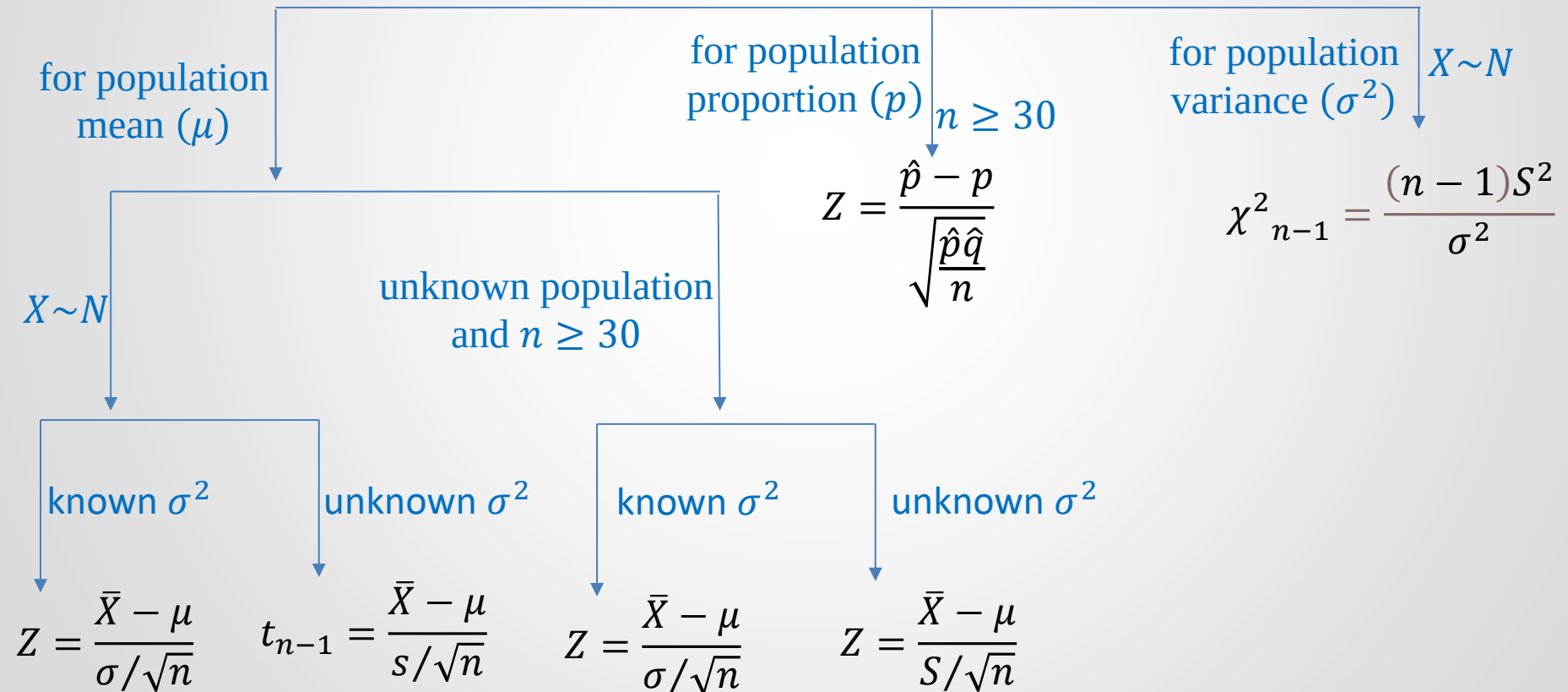
$$\begin{aligned} & P(|\hat{p} - p| < 0.02) \\ &= P\left(\frac{|\hat{p} - p|}{\sqrt{pq/n}} < \frac{0.02}{\sqrt{pq/n}}\right) \\ &= P\left(|z| < \frac{0.02\sqrt{1500}}{\sqrt{\frac{1}{6}\frac{5}{6}}}\right) \\ &= P(|z| < 2.08) \\ &= P(-2.08 < z < 2.08) \\ &= 2(0.4812) = 0.9624 \end{aligned}$$

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OUTLINE: Theorems for a Single Population



Sampling Theory

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In Case of Selection of a Sample without Replacement

When a sample is selected without replacement from a population of size N , and $n/N > 0.05$, the correction factor should be considered. In this case, since

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad \text{or} \quad \sigma_{\bar{x}} = \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \quad \sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \Rightarrow Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \Rightarrow Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

$$t_{n-1} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \Rightarrow t_{n-1} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}\hat{q}}{n}}} \Rightarrow Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}}}$$